

Python Tuples – Quick Reference

1. What is a Tuple?

A **Tuple** is a collection used to store multiple values in a single variable.

```
student = ("Ahmed", 20, "Python")
```

Tuples use **round brackets** ().

2. Tuple vs List ★

List	Tuple
[]	()
Mutable	Immutable
Can be modified	Cannot be modified
Use when data may change	Use when data should remain fixed

Remember:

List → Can Change Tuple → Cannot Change

3. Creating a Tuple

```
numbers = (10, 20, 30, 40)
```

Tuples can contain different data types:

```
student = ("Ahmed", 20, 85.5)
```

4. Accessing Elements

Tuples use indexing, starting from 0.

```
student = ("Ahmed", 20, "Python")
```

```
print(student[0])    # Ahmed
print(student[1])    # 20
```

5. Tuple Cannot Be Modified ★

```
numbers = (10, 20, 30)

numbers[0] = 100    # ✗ Error
```

Tuples are **immutable**.

If you need to add, remove, or change elements, use a **List**.

6. Looping Through a Tuple

```
days = ("Monday", "Tuesday", "Wednesday")

for day in days:
    print(day)
```

7. Useful Functions / Methods

```
numbers = (10, 20, 10, 30)

print(len(numbers))    # Number of elements
print(numbers.count(10)) # Number of occurrences
print(numbers.index(30)) # Position of 30
```

Practical Uses of Tuples

Use Tuples for **fixed groups of related values**:

```
point = (10, 20)        # Coordinates

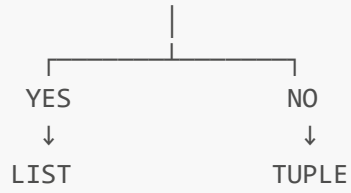
rgb = (255, 0, 0)       # Color

date = (25, 8, 2026)    # Date

days = ("Mon", "Tue", "Wed") # Fixed values
```

Golden Rule

Does the data need to change?



LIST = Mutable TUPLE = Immutable